Ryan Zernach

Senior AI Systems Engineer • LLM Infrastructure & Product Engineering

Portfolio: **Ryan.Zernach.com** • *Email: Ryan@Zernach.com • LinkedIn • GitHub • Remote*

HIGHLIGHTS

- Distributed systems engineer with 10+ years of experience in Payments, Agents, Mobile Apps, MLOps, & AI Infrastructure
- Scaled CellarChat™ tool-calling agent to 1M+ active members, complete with CI/CD LLM evaluation scoring pipelines
- Designed distributed backend systems that support autonomous AI-powered eCommerce VoIP payment workflows
- Completed Gauntlet AI engineering fellowship with 2% acceptance rate and mastered the art surfing the wave of AI
- Boosted mobile app ratings to 4.9★ (10k+ reviews) & lifted recurring revenue by 25%+ w/ Apple Pay subscriptions
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

TECH STACK

Languages
• Python, TypeScript, Rust, Go, Java, C#, Ruby, Swift, Kotlin, Bash, JavaScript, PHP, C++, Objective-C, R, Solidity

AI/ML Systems
• LLMOps, RAG, Agents, MCP, Evaluations, Semantic Kernel, LangChain, ML Pipelines, Computer Vision, Vector Search, Embeddings, Neo4j Graphs, PyTorch, TensorFlow, Scikit-Learn

Platform & Backend Engineering
• Node.js, FastAPI, Nest.js, Express, .NET Core, Microservices, OpenAPI, REST, GraphQL, WebSockets, Webhooks, gRPC, Kafka

Data & Persistence
• PostgreSQL, SQL, NoSQL, Redis, Vector Databases, ETL, Graph Databases, Entity Relationship Modeling, Caching, BigQuery

Cloud Infrastructure & DevOps
• AWS, Azure, GCP, Cloudflare, Docker, GitHub Actions, Terraform, Kubernetes, CI/CD, CloudOps, Monitoring, DNS, Networking

Front-End & Product Engineering
• iOS, Android, Swift, Kotlin, React, React Native, Expo Router, Next.js, Redux, Streamlit, Animations, Design, User Feedback

Distributed Systems & Operations
• Queues, Redis, RabbitMQ, Cron, Analytics, Pub/Sub, Linux, Event-Driven Architecture, VoIP, Observability, Admin Systems, WebRTC, Rate Limiting, Horizontal Scaling, Load Balancing

Security, Identity & Compliance
• AuthN, AuthZ, OAuth, Identity, Role-Based Access Control, Secure Systems Design, SSO, API Security, Secrets, JWT

Payments, Communications & Integrations
• Stripe, Apple Pay, IAP, Billing, Subscriptions, Wallets, Twilio, SMS, Push Notifications, HTTP APIs, Marketing Tech, Email SMTP/IMAP

Blockchain & Web3
• Ethereum, Solana, dApps, WalletConnect, MetaMask, Polygon, Smart Contracts, Solidity, Ethers.js, Tokens, ERC-20, ERC-721

EXPERIENCE

Fractional AI Systems Engineer • *United States • Remote* — ***Jul 2025 - Present***

- Building AI-enabled product and automation systems across payments, mobile apps, and semantic intelligence infrastructure
- Developed semantic intelligence layer around open-source software plugins for managing Plaid financial data and tax filings
- Productionized TranslateWallpaper, a cross-platform mobile app powered by LLM-generated language-learning wallpapers

AI Engineer (Full-Stack) • **CellarTracker** • *Seattle • Remote* — ***Aug 2023 - Jul 2025***

- Scaled agentic CellarChat™ to 1M+ international members with Python back-end & TypeScript front-end
- Architected OpenAPI tool-calling pipeline, fine-tuning pipeline, LLM evaluations, & observability systems
- Integrated intelligent ML-powered recommendations for world's largest online database of wine reviews
- Maintained Microsoft Azure infrastructure & CI/CD pipelines used by international engineering members
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented animations, barcode scanning, activity feed posts, support ticket refactors, & OCR image search
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Migrated legacy support ticket system from Active Server Pages (ASP) to safer, type-friendly Express framework
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and app behavior
- Built contextual AI workflows connecting cellar inventory, tasting history, wine metadata, and recommendations

Mobile Engineer (Full-Stack) • **Archlife Industries Software** • *Miami • Remote* ***Sept 2021 - Aug 2023***

- Delivered eTrading exchange for stocks/crypto, including public leaderboards and copy friend's portfolio distribution
- Engineered companion admin web dashboard to monitor mobile app performance & AWS infrastructure health
- Constructed core mobile app functionality, including: Plaid connections, trending investors, & suggested stocks
- Built social network features, such as direct chat, user profiles, posts, likes, emojis, & threaded comments
- Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto
- Transitioned eSports React Native codebase from JavaScript to TypeScript for long-term maintainability
- Iterated on group chat features, push notifications, & native calendar sync to improve user satisfaction
- Remedied bluetooth connection & map performance for Swift iOS/iPad telecommunications app

Data Science & Machine Learning Tutor • **edX** • *Vanderbilt, UC Berkeley, +24 others • Remote* ***July 2020 – Sept 2021***

- During pandemic, taught 900+ online college students Python, machine learning, TypeScript, websites, & databases
- Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person)
- Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students

React Web Developer (Full-Stack) • **Georgia Hybrid Landscaping** • *Alpharetta, Georgia* ***Oct 2015 – Jan 2019***

- Engineered uploading videos/images, survey questions, pricing proposal, & processing payments
- Migrated websites from PHP to React, and processed $500k+ in gross sales within three years
- Managed online eCommerce payments analytics, Google AdSpend, and email campaigns
- Configured outbound marketing email campaign schedulers & listeners using IMAP/POP
- Setup observability dashboards for payments, marketing, analytics, API logs, & admin

Payments Systems Intern (Card Processing) • **Cost Plus Processing** • *Alpharetta, Georgia* ***Sept 2012 – Jan 2013***

- Worked with merchant processing systems & payment gateways to understand critical financial flows
- Analyzed end-to-end card transaction lifecycles (interchange rates, risk) across payment networks
- Built foundational knowledge of payment infrastructure, interchange fees, & transaction routing

EDUCATION & TRAINING

AI Engineering Fellowship (G3) • **Gauntlet AI** • *Remote • Austin, TX* ***2025***

- Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering
- Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations

Data Science & Machine Learning Bootcamp • **Lambda School** • *Y Combinator • Remote* ***2020***

- Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing
- Python, SciKit-Learn, PySpark, Data Engineering, PostgreSQL Databases, Back-End APIs (Flask / FastAPI)

Turfgrass & Landscape Management • **University of Georgia (NALP)** • *Athens, Georgia* ***2015***

- Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture

Computer Science & Mathematics Coursework • **Georgia Southern University** • *Statesboro, Georgia* ***2012***

- Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression

CERTIFICATIONS

2025 • *Gauntlet AI Engineering Fellowship (G3)*
2024 • *NASA Space Apps Hackathon Finalist*
2022 • *Ethereum Blockchain Developer*
2020 • *TensorFlow Computer Vision*
2020 • *AutoML Pipelines (DataRobot Academy)*
2019 • *Google Cloud Platform GCP Developer*
2019 • *AWS Microservices Cloud Practitioner*
2019 • *Y Combinator Startup School*